Data Analytics Capstone Topic Approval Form

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Capstone Project Name: Logistic Regression on Hotel Reservation Dataset

Project Topic: Logistic Regression Analysis of Hotel Reservations

X This project does not involve human subjects research and is exempt from WGU IRB review.

Research Question: Can Logistic Regression be used to predict the likelihood that a customer will cancel a hotel reservation?

Hypothesis: **Null hypothesis**-. A predictive Logistic Regression cannot be constructed on the research dataset. **Alternate Hypothesis**-. A predictive Logistic Regression model can be constructed on the research dataset with an accuracy greater than 60%.

Context: The contribution of this study to the MSDA program is to employ a Logistic Regression model to help hotels determine which reservations are likely to be cancelled, facilitating better strategic management of their incoming reservations. Hotels across the world have a fundamental business need to ensure that they book as many of their rooms as possible so that they are profitable ventures. However, even reserving every room in a hotel would not suffice to meet this requirement considering that in Europe "in 2018 49.8% of bookings reserved on the OTA Booking.com were cancelled" (Banza, 2020). Banza found during his 2020 CatBoost analysis (a gradient-boosted decision tree) that the larger the lead time on the hotel reservation, the higher the likelihood that the reservation would be cancelled. If approximately half of reservations are canceled, the hotel may want to employ a business strategy of intentionally overbooking to help manage the impact of a high cancellation rate. Since Logistic Regression assesses the probability of a binary outcome based on two or more independent variables, it is a statistically valid strategy for the business problem at hand since the 49.8% cancellation statistic includes the reservation's lead time as well as that it was booked through an online distribution channel (IBM, n.d.).

Data: The dataset for this project was publicly available on Kaggle (Krishnadas, 2022). All-inclusive, the data contains 141,950 records.

The following columns are included in the data for use in the model. The dependent variable (is_canceled) [sic] is also present.

https://www.kaggle.com/datasets/govindkrishnadas/hotel-revenue

Variable	Type
Hotel	Categorical
lead_time	Continuous
arrival_date_year	Categorical
arrival_date_month	Categorical
arrival_date_week_number	Categorical
arrival_date_day_of_month	Categorical
stays_in_weekend_nights	Categorical
stays_in_week_nights	Categorical
Adults	Continuous
Children	Continuous
Babies	Continuous
Meal	Categorical
Country	Categorical
market_segment	Categorical

distribution_channel	Categorical
is_repeated_guest	Categorical (binary)
previous_cancellations	Continuous
previous_bookings_not_canceled	Continuous
reserved_room_type	Categorical
assigned_room_type	Categorical
booking_changes	Continuous
deposit_type	Categorical
days_in_waiting_list	Categorical
customer_type	Categorical
Adr	Continuous
required_car_parking_spaces	Categorical
total_of_special_requests	Continuous
reservation_status_date	Categorical

<u>Limitations</u>: Ideally there would be several other years of data to help establish a trend. It could also have been impactful to see whether those trends changed post-COVID19 by having additional data beyond 2020. There is also no customer demographic data included in the reservation, so customer-centric reservation factors cannot be evaluated as part of this study.

<u>Delimitations</u>: For the sake of limiting regional differences and simplifying the analysis, only the data from Great Britain is used in this project, a delimitation that reduces the record count for the project to 13,487 records. Also, while a variable for Company is present in the data file, it is largely unpopulated and so will not be used as part of the study. Agent will also not be used, as the data does not include a translation for the values of this column that would support meaningful business decision-making. Reservation_status is also highly correlated to is_canceled and so will not be used, as the use of highly correlated independent variables is a potential pitfall for regression analysis (Ranganathan et al, 2017). Record with a reservation_status of "No-Show" will be treated as cancellations since the result is still a vacant room.

Data Gathering: Data will be downloaded by publicly available Excel spreadsheet file from kaggle.com, a 2022 project titled "Hotel Revenue" by Govind Krishnadas. The records will be filtered to a country value of "GBR" to limit to Great Britain. The column for company will be removed. Reservation_status will be used to adjust the is_canceled column for "No-Show" reservations to a 1 (for yes, cancelled) prior to removing the column. This will help to reduce empty variables as well as remove some correlation between independent variables, which can be problematic for regression analyses (Ranganathan et al, 2017). Krishnadas had performed data cleaning on the result set prior to publishing and the data quality is high. It will still, however, be validated that there are not additional data cleaning measures needed to support this specific study. Both the categorical/qualitative and continuous/numeric data will be leveraged in the analysis using Python. Dummy variables will be created via One-Hot Encoding where necessary to support a Logistic Regression model. Overall, data sparsity is less than 1%.

Data Analytics Tools and Techniques: Logistic Regression models do not assume normality or homoscedasticity in the data (Bobbitt, 2020). The data will be evaluated to identify patterns in residuals, to identify and address outliers by imputing to the median, and to ensure sufficient sample size for each variable. Data will be divided into training and testing sets to support model evaluation. The model will be evaluated by ROC Curve, AUC, Accuracy, and AIC. Since the outcome is binary, the accuracy must exceed 50% to perform better than chance. For a business application, accuracy greater than 60% would be preferred. The goal of the study is therefore to create a Logistic Regression model with greater than 60% accuracy and with as few variables as possible. A similar study used Logistic Regression rather than Banza's approach of a CatBoost model and detected correlations between cancelled reservations and the following factors: adults, is repeated guest, previous cancellations, previous bookings not cancelled, booking changes, days in waiting list, required car parking spaces, and total of special requests (Almotiri et al, 2021). This study expects a comparable outcome to those prior findings.

Univariate and bivariate visualizations, Q-Q plots, box plots, and ROC Curves in support of the analysis will be produced with Python.

Justification of Tools/Techniques: As opposed to the "non-standardized code [which] makes R look clunky and awkward," "Python is the go-to language in the production sector because its simple syntax allows programmers to perform complex operations using fewer lines of code" (BasuMallick, 2022). BasuMallik also states that Python is "ranked as one of the most in-demand tech skills by employers, making it an in-demand skill" as opposed to the lower demand for R in non-academic settings. Since the focus of the study is on a business rather than academic question, the more business-oriented tool has been selected for the project – Python. Python also excels at integration into web-based applications: considering that many hotel reservations are made via technology like online travel booking websites, the ability to integrate forecasting with other online datasets or toolsets is an asset to the overall business need. Python's flexibility and therefore option to integrate with other programs or toolsets would provide the business with more downstream options than the stricter limitations of SAS (GeeksForGeeks, 2023).

Project Outcomes: The study will create a Logistic Regression model against hotel reservation data based on a selection of independent variables identified as statistically significant. Banza (2020) has supported that lead time, one of the variables also present in this data set, is a primary indicator of the likelihood of a reservation to result in cancellation, and that outcome would support the alternative hypothesis of this project. Similarly, Almotiri et al (2021) leveraged Logistic Regression to identify correlations between several of the dataset's variables (adults, is repeated guest, previous cancellations, previous bookings not cancelled, booking changes, days in waiting list, required car parking spaces, and total of special requests) and reservation cancellation, also supporting the alternative hypothesis.

Projected Project End Date: September 30, 2024

Sources:

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BasuMallick, Chiradeep. (2022). *R vs. Python: 12 Key Comparisons.* Spiceworks. Retrieved August 7, 2024 from https://www.spiceworks.com/tech/devops/articles/r-vs-python/

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Course Instructor Signature/Date:

☒ The research is exempt from an IRB Review.

☐ An IRB approval is in place (provide proof in appendix B).

Course Instructor's Approval Status: Approved

Date: 8/12/2024

Reviewed by:

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